

LumaPath: An AI/ML-Driven Safety-Aware Route Recommendation System Using Open Geospatial and Real-Time Environmental Data

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Abstract—Conventional navigation systems generally select routes according to distance or estimated travel time, leaving safety information that the user must judge separately. LumaPath addresses this limitation by introducing an AI/ML-based safety analysis stage into the route recommendation. For a selected source and destination, the system obtains multiple candidate routes and examines their geometries using open geospatial and environmental data. Route-level features include travel distance and duration, nearby emergency facilities, and available weather observations; additional features can be incorporated when reliable data become available. These features are processed by an AI/ML safety analysis module to estimate the route safety score and corresponding risk level. The resulting safety assessment is combined with the travel cost to rank the alternatives rather than simply selecting the shortest path. The system uses a React and Leaflet client, FastAPI backend, Project OSRM, OpenStreetMap, Overpass API, Open-Meteo, and PostgreSQL with PostGIS. The design also retains a transparent, rule-based formulation as a baseline for comparison with the learning-based approach. LumaPath is intended to provide a practical and explainable foundation for safety-conscious navigation while allowing the safety model to improve as more representative route data are made available.

Index Terms—AI/ML, safety-aware navigation, route recommendation, machine learning, geospatial analysis, OpenStreetMap, intelligent transportation

I. INTRODUCTION

Navigation applications have made route planning a routine part of daily travel. However, most systems primarily optimize distance, travel time, or a combination of the two. Classical

shortest-path methods provide the foundation for this process, while later work has made route computation efficient for large transportation networks [1], [2]. These objectives are useful for ordinary navigation, but they do not fully represent situations in which a traveller also wants to consider the safety characteristics of a route.

Safety can depend on several factors that are not visible in a conventional route cost. Access to police stations and hospitals, environmental conditions, road characteristics, surrounding activity, and other contextual information can influence how suitable a route is for a particular journey. This is especially relevant for students, people travelling alone, elderly users, tourists, and other travellers who may want safety information before choosing between alternatives. Earlier safe-routing studies have demonstrated that route selection can be changed when safety-related information is incorporated into the planning process [4], [5].

Existing research has approached this problem using crime records, crowdsourced information, geospatial indicators, and machine learning techniques. Some systems use historical incidents directly, while others learn route preferences or risk patterns from available data [6], [7], [12]. Although these approaches demonstrate the value of safety-aware routing, their performance and reproducibility depend strongly on the quality, coverage, and availability of the underlying data.

A closely related multi-factor formulation was proposed by Thilagavathi and Subashini, where several static and dynamic

parameters are combined to estimate route risk [11]. Such a formulation is useful as a transparent baseline, but fixed penalty values do not learn from changing route conditions or from additional observations. This motivates the use of AI/ML in LumaPath. Instead of relying only on manually selected penalties, the proposed system extracts route-level features from available data and uses a learning-based safety analysis stage to estimate the safety of candidate routes.

The objective is not to replace conventional routing, but to add an intelligent safety layer to it. LumaPath first generates practical alternatives, collects information along their actual geometries, prepares the relevant features, and passes them to the AI/ML safety analysis module. The predicted safety information is then combined with distance and travel time so that the final recommendation remains both safety-aware and practical. The use of open geospatial and environmental services also makes the overall workflow suitable for experimentation in regions where proprietary safety datasets are limited.

II. RELATED WORK

Research on safe route recommendations has developed along several related directions. Crime-informed routing uses historical incidents or risk surfaces to modify conventional road costs. Galbrun et al. formulated safe path finding beyond the shortest route, while Levy et al. introduced a learning-based approach that learns safer navigation behavior from urban data [6], [7]. These studies demonstrate that route safety can be treated as a prediction or optimization problem rather than a simple distance calculation.

Other work has applied machine-learning and data-driven methods directly to route risk. Puthige et al. used danger-index calculation and K-means clustering to classify route safety [12]. Zhou et al. combined neural networks with fuzzy evaluation for multi-hazard road risk, illustrating how learning-based models can combine heterogeneous risk indicators [15]. UrbanPathfinders also demonstrates the use of machine learning and geospatial information for safer night-time navigation [14]. Jiang et al. combined multiple sources of roadway information for safe route mapping, further supporting the use of heterogeneous inputs in route-level safety analysis [16].

Participatory systems provide another source of safety information. TREADS used social-media mining and text summarization to produce route-level safety cues, whereas privacy-aware approaches have incorporated crowdsourced observations into route planning [8], [9]. User-specific route recommendations have also been explored for safety-oriented navigation [13]. These systems can capture information that is difficult to obtain from static geographic databases; however, their effectiveness depends on the amount and reliability of user-generated data.

Environmental and infrastructure-based research provides additional evidence for combining multiple route features. Kaur et al. treated safe navigation as a trade-off between safety and route length [3]. Huang et al. investigated route

safety through conflict simulation, while Litzinger et al. examined the use of weather information in route planning [10], [17]. Mukherjee and Mitra studied pedestrian safety at urban intersections, and Al-Bdairi et al. analysed the relationship between weather-related conditions and crash severity [18], [19]. Reviews by Lakshmi and Joseph and by Parvez and Moridpour highlight the broad range of safety factors considered in intelligent transportation research [21], [22].

The main distinction of LumaPath is the placement of AI/ML between data collection and route recommendation. The system uses open data to construct route features and treats the learning model as a component responsible for estimating route safety. The earlier multi-factor formulation is retained as a baseline rather than the final intelligence layer [11]. This separation enables the comparison of fixed rule-based score with a data-driven safety estimate while keeping the route generation and presentation unchanged. OpenStreetMap, Overpass, Project OSRM, and Open-Meteo provide the main data infrastructure [23]–[26]. Recent studies on safer urban navigation using open data also support the practical use of this type of pipeline [27].

III. PROPOSED METHODOLOGY

LumaPath combines route generation, geospatial data collection, feature preparation, AI/ML safety analysis, and route ranking in a single workflow. Figure 1 presents the system architecture. A user supplies the source and destination through the web interface. The backend obtains alternative routes, gathers relevant information for each route, prepares the input features, estimates route safety using the AI/ML module, and returns ranked alternatives to the client.

A. System Architecture

The client is implemented using React, Vite, and Leaflet for location input and map visualization. A FastAPI backend acts as the central orchestration layer. It receives the selected coordinates, communicates with external services, performs route-level processing, invokes the safety analysis module, and returns the results through a JSON-based REST response.

Project OSRM generates alternative routes together with distance, estimated duration, and route geometry. The complete geometry is important because route safety may vary along the path. LumaPath therefore samples locations along each candidate route rather than using only the origin and destination. Overpass is used to obtain nearby hospitals and police facilities, while Open-Meteo supplies available weather observations. PostgreSQL with PostGIS provides storage and caching for route-related information.

B. Route Feature Collection

Each candidate route is represented using its traced geometry and associated travel information. The route is sampled at fixed positions so that the analysis reflects the conditions along the path rather than only at its endpoints. The current prototype uses twelve sampling positions for emergency-service queries and five positions for weather observations. Repeated facilities

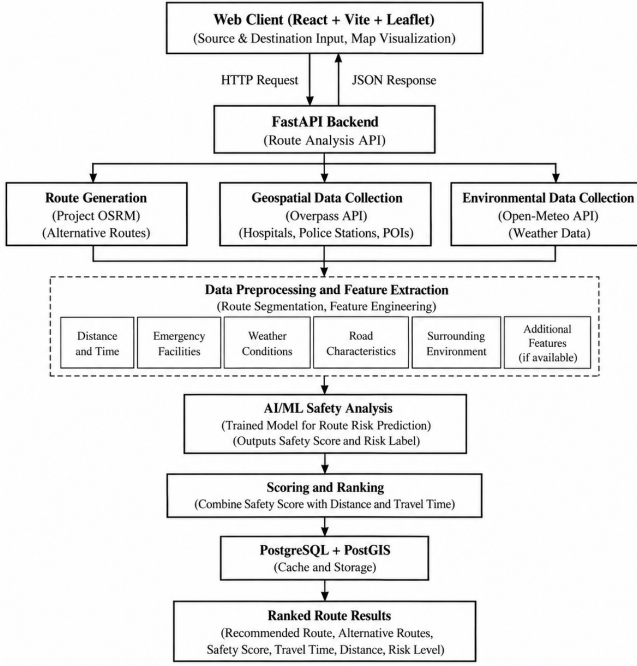


Fig. 1. System architecture.

returned by overlapping search areas are removed before feature calculation.

The present feature set includes route distance, estimated travel time, hospital availability, police-facility availability, precipitation, and wind conditions. These variables were selected because they can be obtained from the current open-data pipeline. The architecture is designed to accept additional features such as traffic, road conditions, lighting, CCTV availability, accident history, time of day, and surrounding activity when reliable data become available. Emergency-service availability has previously been considered a useful indicator for safety-oriented route recommendation, while weather information has been shown to influence route planning [17], [20].

C. AI/ML-Based Safety Analysis

The central addition in LumaPath is an AI/ML module that analyzes the prepared route features instead of relying only on fixed penalty values. For each candidate route, the collected observations are transformed into a consistent feature representation and supplied to the trained safety model. The model estimates the safety level of the route from the combined feature pattern rather than treating every factor as an independently fixed deduction.

This design allows the relationship between route conditions and safety to be learned from the representative training data. The model output was converted into a route-level safety score and corresponding risk category for presentation to the user. The learning stage is separated from the routing engine, which means that the OSRM continues to generate candidate paths, whereas the AI/ML component focuses only on evaluating them. This separation also allows the model to be retrained

TABLE I
ROUTE FEATURES USED BY THE AI/ML SAFETY MODULE

Feature Group	Source	Purpose in Safety Analysis
Distance and duration	Project OSRM	Represents travel cost and route practicality.
Emergency facilities	Overpass/OpenStreetMap	Represents access to nearby assistance.
Weather conditions	Open-Meteo	Represents environmental conditions that affect travel.
Route geometry	Project OSRM	Allows conditions to be analyzed along the complete route.
Additional features	Open data, when available	Extends the model with traffic, roads, lighting, or contextual information.

or replaced without changing the user interface or the route-generation layer.

A rule-based penalty formulation is retained as a baseline because it provides a simple reference against which the learning-based model can be examined. In the baseline, hospitals, police facilities, precipitation, and wind are assigned fixed deductions according to predefined conditions. The purpose of retaining this baseline is not to make it the main recommendation mechanism, but to provide a transparent comparison for the AI/ML output. This is particularly useful when evaluating whether the learning-based approach can capture relationships that are difficult to express through manually selected weights.

The feature representation is intentionally modular. Each route is converted into a common set of values before inference so that the learning model receives comparable inputs for different candidate paths. Normalization and preprocessing are applied consistently between the training and inference stages. This reduces the possibility that differences in numerical scale, such as kilometres versus facility counts, dominate the learned safety estimate.

For model validation, routes should be separated into training and unseen evaluation samples so that the reported performance reflects generalization rather than memorization. The evaluation can be formulated as classification when the model predicts discrete risk levels or as regression when it produces a continuous safety score. In either case, the evaluation should use the final dataset and model configuration adopted by the project. This approach also provides a clear basis for comparing the AI/ML model with the fixed penalty baseline.

D. Safety Score and Route Ranking

The AI/ML module produces the main safety estimate, while the final recommendation also considers travel practicality. A general route objective can be expressed as

$$\Phi_j = \alpha S_j - \beta d_j - \gamma \tau_j, \quad (1)$$

where S_j is the predicted safety score for route j , d_j is its distance, and τ_j is its estimated travel time. The coefficients

TABLE II
ROUTE ANALYSIS PROCEDURE

Step	Operation
1	Receive source and destination coordinates.
2	Request alternative routes from Project OSRM.
3	Sample each route at fixed positions along its geometry.
4	Query Overpass for nearby hospitals and police facilities and remove duplicates.
5	Query Open-Meteo for available weather observations.
6	Construct and normalise the route-level feature vector.
7	Apply the trained AI/ML safety model and obtain the safety score and risk level.
8	Combine safety information with distance and travel time and rank the alternatives.
9	Return the recommended route and supporting route details.

TABLE III
IMPLEMENTATION STACK

Component	Technology
Client	React, Vite, Leaflet
Backend	FastAPI with asynchronous HTTP clients
Base map	OpenStreetMap
Routing	Project OSRM
Geospatial queries	Overpass API
Weather	Open-Meteo
AI/ML analysis	Trained route-safety model
Persistence	PostgreSQL with PostGIS
Interface	JSON over REST

control the relative importance of safety and travel costs. The route with the highest resulting value is presented as the primary recommendation, while the remaining candidates are retained as alternatives.

The model output is intended as decision support rather than a guarantee of real-world safety. The score is influenced by the quality and coverage of the input data and the representativeness of the training data. Presenting the score together with distance, travel time, and the available supporting observations helps users understand the basis of the recommendation.

IV. IMPLEMENTATION

The implementation uses openly accessible components throughout the main data path, as summarized in Table III. The architecture keeps data collection, feature preparation, AI/ML inference, and route presentation as separate stages so that individual components can be updated independently.

The browser sends a single analysis request to the FastAPI service. The backend performs the route and data queries, constructs the feature representation, passes the features to the AI/ML model, and returns the ranked alternatives in a single response. This avoids unnecessary repeated communication between the browser and the backend.

Public geospatial services can be slow or rate-limited. The service uses explicit timeouts and handles incomplete responses without treating missing data as a positive safety signal. The coordinate order is normalized at the service boundary to avoid inconsistencies between mapping components. These

TABLE IV
ROUTE-LEVEL CASE STUDY INPUTS

Route	Dist. (km)	Time (min)	Hosp.	Police
Route 1	8.7	21	6	3
Route 2	9.4	24	4	1
Route 3	11.2	26	1	0

measures improve reliability during repeated demonstrations and testing.

The interface displays the recommended route, along with alternatives. For each route, the user can view distance, estimated duration, safety score, risk level, and available supporting information. This presentation keeps the AI/ML output connected to observable route features instead of exposing only an unexplained numerical recommendation.

V. RESULTS AND DISCUSSION

The current evaluation focuses on the route-analysis pipeline and the behavior of the safety assessment on urban and suburban roads in Kerala, India. A representative source and destination pair separated by approximately nine kilometers produced three structurally different alternatives through Project OSRM. The same data-collection session was used for the alternatives so that the comparison was based on a common set of environmental observations.

The case study illustrates why route-level safety analysis requires more than distance alone. Route 1 has the strongest emergency service coverage and the lowest travel cost among the alternatives. Route 2 is slightly longer and has fewer nearby facilities. Route 3 is the longest route and has no police facility within the selected search areas. These differences provide distinct feature patterns for the AI/ML safety analysis stage.

The main purpose of the learning-based formulation is to move beyond manually assigning a fixed penalty to each observation. Instead, the model receives the combined route features and estimates their relationship with route safety. During validation, the AI/ML module should therefore be assessed using held-out route samples and appropriate classification or regression measures depending on the final model formulation. Accuracy, precision, recall, F1-score, mean absolute error, or related measures should only be reported when they are calculated from the actual evaluation dataset.

The rule-based formulation is useful as a baseline because its behavior can be traced directly to individual conditions. The AI/ML approach is intended to provide a more flexible representation of the same safety problem. A meaningful comparison should therefore examine not only the numerical score but also whether the model produces consistent route ordering, responds to changes in input conditions, and generalizes beyond the routes used during training.

The open-data design also introduces limitations. OpenStreetMap coverage can vary between locations, and the absence of a mapped facility does not necessarily mean that the facility does not exist. Weather observations are also dependent

TABLE V
COMPARISON OF THE BASELINE AND PROPOSED SAFETY ANALYSIS

Aspect	Rule-Based Baseline	LumaPath AI/ML Approach
Safety representation	Fixed penalties	Learned relationship between route features and safety
Feature contribution	Manually specified	Determined through model training
Adaptability	Limited to predefined rules	Can be updated by re-training with representative data
Complex relationships	Difficult to represent explicitly	Can model interactions among multiple features
Interpretability	Direct and transparent	Requires supporting feature information and validation
Primary role	Reference baseline	Main safety-analysis component

on the resolution and update frequency of the external service. These limitations affect both the rule-based baseline and the AI/ML model because the model cannot learn information that is not represented in its inputs.

Another important limitation is the availability of representative training data. A safety model trained on a narrow geographic area may not generalize to other cities or road environments. The training dataset should therefore contain sufficient variation in route types, environmental conditions, infrastructure availability, and travel contexts. Future evaluation should include multiple locations and time periods before the predicted score is treated as a broadly applicable safety indicator.

Therefore, the current system is positioned as an intelligent decision-support tool rather than an absolute safety predictor. The AI/ML module provides an additional analytical layer, while the interface continues to expose alternative routes and supporting information so that users can consider the recommendations in context.

The comparison in Table V highlights the reason for introducing the learning-based component. A fixed rule can be useful when the relationship between an observation and perceived risk is already known; however, it becomes difficult to maintain as the number of features increases. An AI/ML model provides a mechanism for learning from combined route characteristics and can therefore represent interactions that are not easily expressed as independent deductions. At the same time, the proposed system does not remove the baseline because a transparent reference is useful when analyzing model behavior.

The architecture also separates prediction from recommendation. The AI/ML model is responsible for estimating route safety, whereas the final ranking stage considers safety along with distance and travel time. This separation is important because a highly safety-oriented prediction should not automatically produce an impractical route. It also allows the model to be improved independently of the route-generation service and map interface.

For a complete experimental evaluation, future experiments should report the exact model configuration, training-set size, feature preprocessing, validation strategy, and performance measures used by the final implementation. Results should be reported on routes that were not used during training, and comparisons should include the rule-based baseline. Such reporting would allow the contribution of the AI/ML component to be evaluated independently from the underlying mapping and routing services.

VI. CONCLUSION AND FUTURE SCOPE

LumaPath presents an AI/ML-driven approach to safety-aware route recommendation using open geospatial and environmental information. The system generates alternative routes, extracts route-level features, applies an AI/ML safety analysis model, and combines the resulting safety estimate with travel cost for route ranking. A transparent rule-based formulation is retained as a baseline, allowing the learning-based approach to be evaluated against a simpler existing strategy.

Future work will focus on strengthening the training dataset, validating the model across different locations and travel conditions, and incorporating additional features such as traffic, road condition, lighting, CCTV availability, accident history, time of day, and user feedback where reliable data can be obtained. These additions can improve the ability of LumaPath to provide context-sensitive safety information while maintaining a modular and explainable architecture.

REFERENCES

- [1] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, no. 1, pp. 269–271, 1959.
- [2] H. Bast, D. Delling, A. Goldberg, M. Muller-Hannemann, T. Pajor, P. Sanders, D. Wagner, and R. F. Werneck, "Route planning in transportation networks," in *Algorithm Engineering*, LNCS, vol. 9220, Springer, 2016, pp. 19–80.
- [3] R. Kaur, V. Goyal, V. M. V. Gunturi, A. Saini, K. Sanadhya, R. Gupta, and S. Ratra, "A navigation system for safe routing," in *Proc. IEEE Int. Conf. Mobile Data Management*, 2020, pp. 1–10.
- [4] D. Bura, M. Singh, and P. Nandal, "Predicting secure and safe route for women using Google Maps," in *Proc. COMITCon*, 2019, pp. 103–108.
- [5] A. Eranpurwala, F. Indorewala, N. Mapari, and S. Mishra, "Women safety application for safe route prediction," *Int. Research J. Engineering and Technology*, vol. 8, no. 5, pp. 2278–2282, 2021.
- [6] E. Galbrun, K. Pelechris, and E. Terzi, "Urban navigation beyond shortest route: The case of safe paths," *Information Systems*, vol. 57, pp. 160–171, 2016.
- [7] S. Levy, W. Xiong, E. Belding, and W. Y. Wang, "SafeRoute: Learning to navigate streets safely in an urban environment," *ACM Trans. Intelligent Systems and Technology*, vol. 11, no. 6, pp. 1–17, 2020.
- [8] F. T. Islam, T. Hashem, and R. Shahriyar, "A privacy-enhanced and personalized safe route planner with crowdsourced data and computation," in *Proc. IEEE Int. Conf. Data Engineering*, 2021, pp. 229–240.
- [9] K. Fu, Y. Lu, and C. Lu, "TREADS: A safe route recommender using social media mining and text summarization," in *Proc. ACM SIGSPATIAL Int. Conf. Advances in Geographic Information Systems*, 2014, pp. 557–560.
- [10] H. Huang, Y. Wei, C. Han, J. Lee, S. Mao, and F. Gao, "Travel route safety estimation based on conflict simulation," *Accident Analysis and Prevention*, vol. 171, 106666, 2022.
- [11] T. Thilagavathi and A. Subashini, "Multi-factor risk assessment and route optimization for safe human travel," *Int. J. Advanced Computer Science and Applications*, vol. 15, no. 11, pp. 426–435, 2024.

- [12] I. Puthige, K. Bansal, C. Bindra, M. Kapur, D. Singh, V. K. Mishra, A. Aggarwal, J. Lee, B.-G. Kang, Y. Nam, and R. R. Mostafa, "Safest route detection via danger index calculation and K-means clustering," *Computers, Materials and Continua*, vol. 69, no. 2, pp. 2761–2777, 2021.
- [13] Y. S. Asawa, S. R. Gupta, V. Vaishnavi, and N. J. Jain, "User specific safe route recommendation system," *Int. J. Engineering Research and Technology*, vol. 9, no. 10, pp. 574–580, 2020.
- [14] E.-C. Toda-Puiulet and A.-I. Trifan, "Safest paths for women at night in Cluj-Napoca: A feature offered by UrbanPathfinders," in *Proc. ACM Celebration of Women in Computing*, 2025.
- [15] C. Zhou, M. Chen, J. Chen, Y. Chen, and W. Chen, "A multi-hazard risk assessment model for a road network based on neural networks and fuzzy comprehensive evaluation," *Sustainability*, vol. 16, no. 6, 2429, 2024.
- [16] S. Jiang, M. Jafari, M. Kharbeche, M. Jalayer, and K. N. Al-Khalifa, "Safe route mapping of roadways using multiple sourced data," *IEEE Trans. Intelligent Transportation Systems*, vol. 23, no. 4, pp. 3169–3179, 2022.
- [17] P. Litzinger, G. Navratil, A. Sivertun, and D. Meier, "Using weather information to improve route planning," in *Lecture Notes in Geoinformation and Cartography*, Springer, 2012, pp. 199–214.
- [18] D. Mukherjee and S. Mitra, "Pedestrian safety analysis of urban intersections in Kolkata, India using a combined proactive and reactive approach," *J. Transportation Safety and Security*, vol. 14, no. 5, pp. 754–795, 2020.
- [19] N. S. S. Al-Bdairi, S. L. Zubaidi, H. Zubaidi, and I. Obaid, "Injury severity of single-vehicle weather-related crashes on two-lane highways," *J. Transportation Safety and Security*, pp. 1–21, 2023.
- [20] K. Pawoskar and R. Kumar P., "Safest route detection application," *Int. Research J. Engineering and Technology*, vol. 7, no. 5, pp. 5210–5214, 2020.
- [21] A. V. Lakshmi and K. S. Joseph, "Travel safe: A systematic review on safe route guidance system," in *Proc. IEEE Conf. Interdisciplinary Approaches in Technology and Management for Social Innovation*, 2022, pp. 1–6.
- [22] M. S. Parvez and S. Moridpour, "Application of smart technologies in safety of vulnerable road users: A review," *Int. J. Transportation Science and Technology*, 2024.
- [23] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," *IEEE Pervasive Computing*, vol. 7, no. 4, pp. 12–18, 2008.
- [24] R. Olbricht, "Overpass API: A database engine for OpenStreetMap data," OpenStreetMap Foundation, Technical Documentation, 2024.
- [25] D. Luxen and C. Vetter, "Real-time routing with OpenStreetMap data," in *Proc. ACM SIGSPATIAL Int. Conf. Advances in Geographic Information Systems*, 2011, pp. 513–516.
- [26] P. Zippenfenig, "Open-Meteo.com weather API," Zenodo, 2023, doi: 10.5281/zenodo.7970649.
- [27] V. Yashodha, S. Pagadala, C. S. Bhavani, S. Kamatchi, and J. M. Oli, "Rethinking urban navigation for safer journeys," in *Proc. Int. Conf. Emerging Technologies in Computing and Communication*, IEEE, 2025.